

13 Bus Power System Study

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1 THE SYSTEM AND WHAT I TESTED

I built a 13-bus power system in PowerWorld to see how it handled changes in demand and the loss of major equipment. It had two generators, transmission lines, step-down transformers, and five load buses. I wanted to get the system running within the limits I chose, increase the load by 15%, and then see which outages caused the most trouble. One pattern kept showing up: the voltage could drop too low even when the lines and transformers were still below their loading limits. That became the main finding of the study.

The model used three nominal voltage levels. Buses 1, 2, 3, 7, and 8 made up the 138 kV transmission network. Buses 4, 5, 6, 9, 10, and 11 operated at 69 kV, while Buses 12 and 13 operated at 13.8 kV. Generator 1 was connected to Bus 1 and served as the slack generator, adjusting its output to balance demand and losses. Generator 2 was connected to Bus 7 and was set to supply 100 MW. Its reactive-power output changed as needed to help support the voltage.

I used parallel circuits between Buses 1 and 2 and between Buses 2 and 3. Another route through Buses 2, 7, 8, and 3 gave power an extra path through the transmission network. Two 138/69 kV transformers, one between Buses 3 and 4 and the other between Buses 8 and 9, supplied the lower-voltage sections. The 69 kV network included the 4–5–6 loop, the 4–9 tie, and the 4–10–11–9 path. Transformers from Bus 10 to Bus 12 and from Bus 11 to Bus 13 supplied the distribution loads. Figure 1.1 shows how these parts fit together.

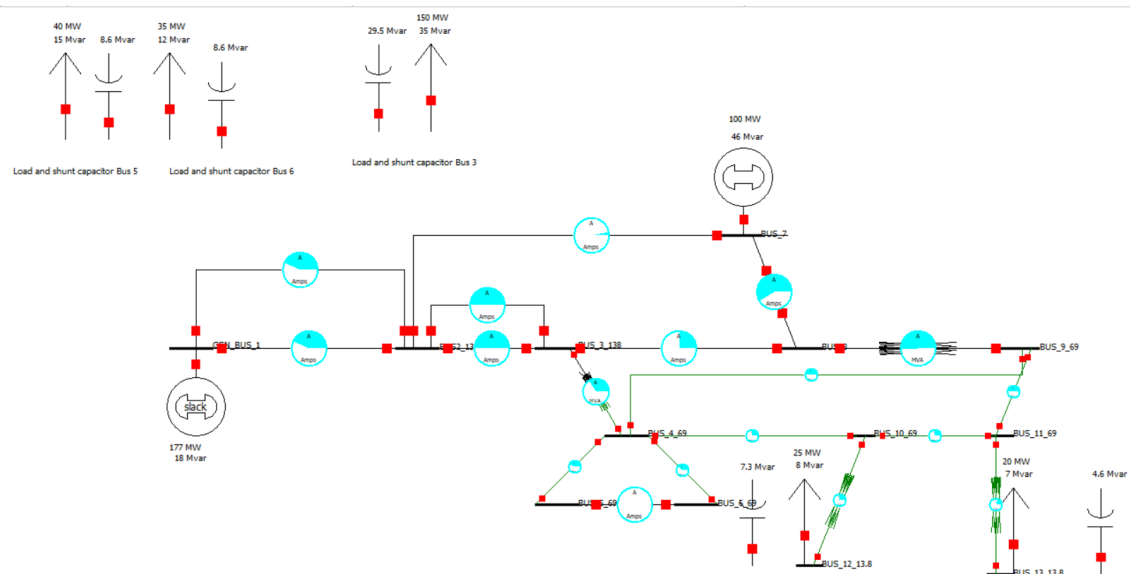


Figure 1.1: The completed 13-bus network after voltage adjustments, with the original 270 MW load. The generator and shunt outputs shown on the diagram are rounded.

For voltage, I used an acceptable range of 0.95 to 1.05 pu. For branch loading, I used 100% of the monitored equipment rating as the limit. These project limits aren't a complete utility planning standard. I checked voltage and loading separately because a line or transformer can stay within its rating while the voltage farther downstream is too low. Most generator readings were rounded to whole MW and Mvar values, so the losses calculated from those readings are approximate too.

I kept the original case and the load-growth case separate when comparing results. The original 13-bus system had 270 MW of demand. After the 15% increase, it had 310.5 MW. For the final outage tests, I used the load-growth case with the added voltage support as my normal operating reference. Comparing an outage at 310.5 MW against the original 270 MW case would mix together the effects of the outage and the extra load.

2 BUILDING THE BASE CASE AND IMPROVING VOLTAGE

When I expanded the network to 13 buses, PowerWorld could solve the power flow, but several voltages were still too low. Bus 9 was at about 0.925 pu, Bus 10 at 0.897 pu, and Bus 11 at 0.906 pu. The voltage dropped further downstream: Bus 12 was at approximately 0.888 pu, and Bus 13 was at approximately 0.899 pu. All five were below my 0.95 pu minimum. Bus 12 had the lowest voltage.

The loading results helped me decide what to work on. In this initial case, the Bus 10–12 transformer was only about 26.5% loaded, and the Bus 11–13 transformer was about 21.4% loaded. Both had plenty of unused capacity, even though their receiving-end voltages were low. I focused on voltage regulation and reactive-power support before considering larger transformers. The Bus 10–11 tie also carried about 12.9 MW from Bus 11 toward Bus 10. That showed the alternate route was already helping supply the load.

Adding the second 138/69 kV transformer helped share the downstream demand between

two supply points. During the earlier 225 MW stage, the Bus 8–9 transformer received 39.88 MW and 18.49 Mvar at its sending end. It delivered 39.78 MW and 16.91 Mvar at its receiving end. PowerWorld showed losses of 0.099 MW and 1.580 Mvar. Subtracting the rounded real-power readings gives 0.10 MW, which agrees with the displayed loss once rounding is taken into account.

The sending-end apparent power was 43.96 MVA, or 21.98% of the transformer's 200 MVA rating. At the receiving end, it was 43.23 MVA. The difference makes sense because some power is lost in the transformer. At this stage, the original Bus 3–4 transformer supplied about 36.1 MW. Together, the two transformers supplied approximately 76 MW for the 75 MW lower-voltage load and the network losses. These readings came from an earlier stage of the model, so I kept them separate from the final 270 MW results.

I used shunt capacitors and transformer tap changes to improve voltage in the completed system. The capacitors supplied reactive power at the buses where they were connected, reducing how much had to come through the upstream network. The transformer taps changed the effective voltage ratio. Both helped, but in different ways. A tap change could raise the downstream voltage; it didn't create extra real power or make the load demand disappear.

The last base-case adjustment was at Bus 12. I increased its capacitor setting from 7.5 Mvar to 8 Mvar, but the voltage in the Bus 10–12 section was still too low. The capacitor change wasn't enough on its own. I then changed the Bus 10–12 transformer tap from 1.00 to 0.99, which gave the voltage the additional increase it needed. This worked for my transformer's connection. I wouldn't assume the same effect with every transformer connection.

After these changes, the recorded base-case voltages met the lower limit I had chosen. Generator 1 supplied approximately 177 MW and 18 Mvar, while Generator 2 supplied 100 MW and 46 Mvar. That gave a total of about 277 MW of generation for 270 MW of load. The estimated real-power loss was $277 - 270 = 7$ MW, or about 2.6% of the load. This became my starting point for the load-growth study.

3 INCREASING THE LOAD BY 15 PERCENT

For the growth study, I increased both the real and reactive demand at every load bus by 15%. Since I increased them by the same factor, each load kept the same power factor. Table 3.1 lists the exact settings. Total demand went from 270 MW and 77 Mvar to 310.5 MW and 88.55 Mvar. That's an increase of 40.5 MW and 11.55 Mvar. I calculated these totals from the load settings because the rounded labels on the one-line diagram don't show all the decimal places.

Table 3.1: Load settings before and after the 15% increase.

Load bus	Original MW	Original Mvar	Growth MW	Growth Mvar
3	150	35	172.50	40.25
5	40	15	46.00	17.25
6	35	12	40.25	13.80
12	25	8	28.75	9.20
13	20	7	23.00	8.05
Total	270	77	310.50	88.55

Generator 2 stayed at its 100 MW setting, so the slack generator had to pick up the extra demand and losses. In the first growth run, Generator 1 rose to approximately 220 MW and 32 Mvar, while Generator 2 supplied approximately 71 Mvar. Total real-power generation was about 320 MW. Subtracting the 310.5 MW load gives an estimated loss of 9.5 MW, or approximately 3.1% of demand. The higher losses made sense with more power and current flowing through the network.

The power flow solved, and the recorded equipment loading stayed below the thermal limit. The voltages still needed work, though. Bus 11 fell to about 0.94 pu, and Bus 13 fell to about 0.93 pu. The generators could supply the demand, but that alone didn't make the case acceptable. Based on these results, the next step was to improve voltage support. The recorded loading didn't point to an immediate need for higher-rated lines or transformers.

The capacitor output also changed with bus voltage. In PowerWorld, actual shunt Mvar equals the nominal setting multiplied by the square of the bus voltage in per unit [1]. For example, a nominal 5 Mvar bank at 0.93 pu would supply $5 \times 0.93^2 = 4.3245$ Mvar, or approximately 4.32 Mvar. I calculated this example to explain the behavior; it isn't a separate simulation result. It shows why a capacitor can supply less reactive power than its nominal rating just when the bus voltage is already low.

I added switched shunt support at Bus 4 and then at Bus 10 because the low-voltage problem extended into the 69 kV network. Supporting only the distribution buses hadn't been enough. The Bus 10 bank could switch its output in discrete steps to help control voltage. The available shunt blocks and the chosen voltage band determine those changes [2]. The bank could then adjust local reactive support as conditions changed.

With the extra support in place, Generator 1 supplied approximately 220 MW and 25 Mvar, while Generator 2 supplied 100 MW and 50 Mvar. Their combined reactive output had dropped from about 103 Mvar in the first growth run to 75 Mvar. The load still needed 88.55 Mvar; the local shunts were now supplying part of it. Bus 4 was at about 0.958 pu, and Buses 5 and 6 were at about 0.951 pu. Those values passed the lower limit, but only just at the latter buses. They were 0.001 pu above it, so there wasn't much room for another voltage drop.

4 WHAT HAPPENED DURING EQUIPMENT OUTAGES

Next, I tested outages of Generator 2, each of the two main step-down transformers, and the Bus 3–8 transmission line, one at a time. I wanted to see whether the remaining equipment could still supply the load while keeping voltage and loading within the chosen limits.

Table 4.1 compares the recorded results at the higher load level. Where my voltage notes referred to a group of buses, I kept that area description instead of guessing an exact location.

Table 4.1: Recorded voltage and branch-loading results for the supported growth case and individual outages.

Operating case	Lowest recorded voltage	Largest recorded branch loading	Screening result
Healthy growth reference	0.951 pu Buses 5 and 6	58.32% MVA Bus 2–3 circuit	Acceptable recorded values
Generator 2 out	0.86621 pu Bus 4–6 area	89.27% Bus 1–2 circuit	Low voltage
Bus 3–4 transformer out	About 0.96 pu Bus 8–9 area	72.95% Bus 7–8 line	Acceptable recorded values
Bus 8–9 transformer out	0.8382 pu Bus 4–6 area	72.50% Bus 3–4 transformer	Low voltage
Bus 3–8 line out	0.9470 pu Bus 4–6 area	59.82% Bus 7–8 line	Slightly low voltage

Table 4.1 uses the loading percentages recorded during the study. The normal operating case specifically recorded MVA loading. Some outage notes didn't say whether their percentages were based on MVA or current, so I haven't treated those as exactly the same measurement. None of the listed percentages went above 100%. Even so, several cases failed because the voltage dropped below the chosen minimum. Checking loading by itself would have missed those problems.

Generator outage. When I took Generator 2 out of service, Generator 1 had to supply more real power and provide more reactive support. In the final outage record, Generator 1 supplied approximately 325 MW and 148 Mvar, compared with 220 MW and 25 Mvar in the normal operating case. Estimated losses were $325 - 310.5 = 14.5$ MW. Loading on the Bus 1–2 circuit reached 89.27%, and the lowest recorded voltage in the Bus 4–6 area was 0.86621 pu. The load was still supplied in the power-flow solution, but the voltage was well below the acceptable range.

Another screenshot of a generator outage showed approximately 326 MW and 149 Mvar from Generator 1, with the two Bus 1–2 current indicators near 88%. I kept these readings separate because they came from a different operating snapshot. Both showed the same general result: losing the second generator made the slack generator work much harder and reduced voltage support. The loading was higher, but an 88% or 89.27% reading was still below the 100% limit. Calling it an overload would be incorrect.

In a separate attempt to improve the generator-outage case, a transformer tap was changed to 0.95. The screenshot after the change showed Bus 4 at 0.986814 pu, or 68.090 kV on its 69 kV base. Its switched shunt supplied 9.738 Mvar from a nominal 10 Mvar setting, which fits the voltage-squared relationship. Bus 4 had clearly improved. I'd still need to check the other buses, branches, and generator limits before calling the whole corrected case acceptable. Also, just because PowerWorld finds a slack-generator output doesn't mean the physical generator has enough capacity or reserve to keep supplying it.

Transformer outages. When I removed the Bus 3–4 transformer, the other substation

supplied the network through an alternate route. The recorded voltage in the Bus 8–9 area was about 0.96 pu, and the highest recorded loading was 72.95% on the Bus 7–8 line. Generator 1 supplied approximately 221 MW and 35 Mvar, while Generator 2 supplied 100 MW and 59 Mvar. Estimated losses were about 10.5 MW. This case passed the recorded voltage and loading checks. It showed how the second transformer and the connections within the 69 kV network helped during an outage.

The result was worse when I removed the Bus 8–9 transformer. Voltage in the Bus 4–6 area fell to 0.8382 pu, even though the Bus 3–4 transformer was only 72.50% loaded. Generator 1 supplied about 221 MW and 52 Mvar, and Generator 2 supplied 100 MW and 56 Mvar. Estimated real-power losses were again about 10.5 MW. The two transformer outages clearly didn't affect the system in the same way. What mattered was where power could still enter the network, which paths it could follow, and how much reactive support remained.

Transmission line outage. Opening the Bus 3–8 line sent power through the remaining paths in the transmission network. Generator 1 supplied approximately 220 MW and 16 Mvar, while Generator 2 supplied 100 MW and 58 Mvar. The highest recorded loading was 59.82% on the Bus 7–8 line. Voltage in the Bus 4–6 area fell to 0.9470 pu, which was 0.003 pu below the chosen minimum. It was a small miss, but it still failed the voltage check. The drop was less severe than in the generator and Bus 8–9 transformer outage cases.

These tests showed that having alternate paths helped keep the network connected, but some outages still left it with low voltage. Losing the Bus 8–9 transformer gave the lowest recorded voltage. Losing the generator caused the highest recorded loading and required the most output from the slack generator. I'd focus on those cases first, adjust voltage support, and check all the limits again.

5 THE THREE PHASE FAULT AND WHAT I FOUND

I also tested a balanced three-phase fault at Bus 8, which operated at 138 kV. I set both fault resistance and reactance to zero to represent a bolted fault. PowerWorld showed an overall fault-current magnitude of 1540.540 A, or approximately 1.54 kA. The three phase currents were nearly equal, with angles of -47.50° , -167.50° , and 72.50° . Those angles were separated by 120° , as expected for a balanced fault. Figure 5.1 shows the fault settings and the calculated subtransient phase currents.

Fault Location		Fault Type	
☒ Bus Fault	☐ In-Line Fault	☐ Single Line-to-Ground	☒ 3 Phase Balanced
		☐ Line-to-Line	☐ Double Line-to-Ground

Fault Impedance		Fault Current		Subtransient Phase Current	
R : 0.00000	X : 0.00000	Scale Current By: 1.00000		Amps	deg.
		If Magnitude: 1540.540	Amps	A 1540.540	-47.50
		If Scaled Mag: 1540.540	Amps	B 1540.500	-167.50
		If Angle: -47.50	deg.	C 1540.500	72.50

Units	
☐ p.u.	☒ Amps

Figure 5.1: The Bus 8 balanced three-phase fault settings and calculated subtransient currents.

At the faulted Bus 8, voltage dropped to 0.00000 pu in all three phases. The effect spread through the rest of the network too. The fault-results table showed approximately 0.31338 pu at Bus 1, 0.03352 pu at Bus 9, 0.05027 pu at Bus 12, and 0.04198 pu at Bus 13. These readings describe the system during the fault, so they shouldn't be compared directly with the post-outage operating voltages in Table 4.1. They show how strongly a fault on the transmission side can affect loads much farther downstream.

The fault-current result depends on the data used in this model. PowerWorld uses device fault data and can fall back on load-flow defaults when separate fault data aren't available [3]. Before using the result to choose breakers or coordinate protection, I'd check the generator subtransient data and the rest of the network fault model. I didn't test breaker clearing times, relay operation, or transient stability here. The fault current alone can't show how the system would behave over time.

By the end of the study, I'd built the 13-bus network, improved the base-case voltages, and supplied the 15% higher load with added reactive support. The original case supplied 270 MW with approximately 7 MW of losses. At 310.5 MW of demand, the supported system had approximately 9.5 MW of losses and passed the recorded voltage checks under normal conditions. Some buses were still close to the lower limit, though, and the outage tests showed why that small margin mattered.

The system still struggled with several outages. Losing Generator 2 or the Bus 8–9 transformer caused a large voltage drop, while losing the Bus 3–8 line caused a smaller one. Because of that, I wouldn't call the final network fully N–1 secure. My next step would be to adjust the transformer taps and switched shunts for the weakest cases, check the generators' capability limits, and run the voltage and loading checks again. The main point I took from the study was that a system can have enough MW supply and unused equipment capacity and still need better voltage support.

REFERENCES

- [1] PowerWorld Corporation. *Switched Shunt Information in Run Mode*.
- [2] PowerWorld Corporation. *Switched Shunt Control*.
- [3] PowerWorld Corporation. *Fault Analysis Dialog*.